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Household practices associated with reported household Lassa fever occurrence in hyperendemic communities of Edo state, Nigeria

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Abstract

Background Lassa fever (LF) remains a major public health threat in Nigeria, with Edo State designated a hyperendemic hotspot. Despite high community awareness and ongoing public health interventions, households continue to report LF annually, suggesting that household-level practices may be associated with the observed disease patterns. This study investigated household practices associated with self-reported prior household LF occurrence in Ekpoma communities, Edo State.

Methods An analytical cross-sectional study was conducted among household heads or representatives selected through multistage sampling between June and August 2023. Data on socio-demographics, LF knowledge, and household practices were collected using an interviewer-administered questionnaire. Associations between household practices and self-reported household LF occurrence within 12 months preceding the study were assessed using the chi-square test and multivariate logistic regression. The level of significance was set at $P < 0.05$.

Results Among 303 respondents recruited for the study, more than half were male (58.1%) and middle-aged (mean age: 44.6 ± 17.8 years). Farmers (26.4%) and entrepreneurs (26.1%) were the largest occupational groups. Although 92.4% reported LF awareness and 76.6% had received community sensitisation, most households reported rodent presence (93.7%). Furthermore, 53.8% stored food in open containers and 19.1% stored food on the floor. Multivariate analysis showed that storing food on the floor (AOR = 2.40; 95% CI: 1.12–5.19) and storing food in uncovered containers (AOR = 5.16; 95% CI: 1.23–21.71) were associated with significantly higher odds of reported prior household LF occurrence.

Conclusion The findings highlight a potential knowledge-to-practice gap in Lassa fever prevention in hyperendemic communities. Household food storage practices, particularly food storage on the floor and in uncovered containers, were associated with reported prior household LF occurrence, underscoring the need to strengthen community-based food storage and environmental sanitation interventions.

Keywords Household Lassa fever occurrence, Household practices, Rodent control, Endemic communities, Nigeria

1 Introduction

Lassa fever (LF) is a zoonotic, acute viral hemorrhagic illness caused by the Lassa virus (LASV), an enveloped, single-stranded bisegmented RNA virus belonging to the family *Arenaviridae* [1–4]. Since its clinical discovery in 1969 in Lassa village, Borno State, the disease has evolved from an occasional medical curiosity into a permanent public health crisis in West Africa [5, 6]. The World Health Organization (WHO) currently lists LF as a priority pathogen requiring urgent research and development due to its high case-fatality rate (CFR) and the absence of a licensed vaccine [7]. In Nigeria, the epidemiological profile of the disease has shifted from seasonal outbreaks to a perennial occurrence, with the Nigeria Centre for Disease Control reporting several confirmed cases and deaths annually [8–15]. As of epidemiological week 50, 2025, Nigeria has reported 1,097 confirmed LF cases with 201 deaths (CFR 18.3%) across 21 states and 103 Local Government Areas (LGAs), with Ondo, Bauchi, Edo, and Taraba accounting for 89% of cases.

Edo State in Southern Nigeria is one of the national epicentres for LF, consistently accounting for a disproportionate share of the annual national burden and deaths from LF, particularly within the state's northern senatorial district, which comprises administrative hubs such as Ekpoma and Irrua [16–23]. The hyperendemicity of this region has been linked to a complex interplay of ecological factors and human behaviour that facilitate the continuous spillover of the virus from its primary reservoir, the "multimammate rat" (*Mastomys natalensis*), to human populations [9, 11, 24, 25]. Unlike many other rodents, *Mastomys* species are highly synanthropic, as they thrive in and around human dwellings, utilising domestic spaces for food and shelter [26–29].

The transmission dynamics of LF are primarily driven by household-level interactions [11, 30, 31]. Direct transmission occurs through contact with rodent excreta (urine and faeces), through contaminated surfaces or the ingestion of contaminated food materials [11, 30, 31]. Despite extensive public health campaigns, hyperendemic communities in Edo State exhibit a persistent knowledge-practice gap [11, 32]. While community members often possess high levels of awareness regarding the symptoms and dangers of the disease, their adherence to preventive household practices remains suboptimal [11, 32, 33]. This disconnect suggests that household practices and behavioural drivers are not merely a result of ignorance but are deeply embedded in socioeconomic realities and traditional lifestyles [11, 34].

Some of the reported household practices associated with sustained transmission include the improper storage of food items [11, 34]. In rural Edo State, many households store grains and flour in open containers or on the floor, providing easy access to rodents [11, 34]. Furthermore, the common practice of spread-drying agricultural produce, such as cassava flakes (garri) and grains, along roadsides and open spaces exposes these food sources to rodent activity [11, 34]. This environmental interface is a critical point of viral spillover, as the LASV can remain infectious in the environment for several days under specific humidity and temperature conditions [11, 16, 34–36].

Cultural factors also play a pivotal role in sustaining the LF transmission cycle. The hunting, processing, and consumption of rodents as a source of animal protein remain prevalent in certain sub-populations, leading to direct blood-borne exposure during butchering [11, 37–39]. Additionally, poor domestic hygiene and inadequate waste management systems within the household create harborage sites where piles of refuse or cluttered storage areas encourage rodent nesting [11, 37–39]. These practices are often necessitated by the lack of modern infrastructure, making behaviours that increase LF risk a consequence of structural poverty [40, 41].

The persistence of LF in hyperendemic communities suggests that top-down, generalised public health messaging may be insufficient [42–45]. Prevention strategies often focus on clinical management, including case isolation and large-scale vector control, yet the most effective barrier against the disease is located at the household level [42, 43]. To break the cycle of transmission, there is an urgent need to identify the specific, ingrained household practices that sustain the virus [11, 42, 43, 45]. This requires a granular understanding of the domestic practices, including how food is stored, how waste is disposed of, and how residents interact with their immediate surroundings [11, 42, 43]. Understanding these practices is essential for developing targeted community-level prevention strategies and interventions that are not only scientifically sound but also culturally acceptable and economically feasible for the local population [46].

This study investigated the household practices that are associated with self-reported household LF occurrence in hyperendemic communities of Edo State. By investigating the intersection of sanitation, food security, and human-rodent interaction, the research aims to provide a baseline for evidence-based, community-led interventions for future prevention of LF outbreaks in Nigeria.

2 Methods

2.1 Study design

An analytical cross-sectional study design was employed to obtain quantitative data, adopting the methods utilised in the study by Bonwitt et al. [47].

3 Study area

The study was conducted in Ekpoma, the administrative headquarters of the Esan West LGA, Edo state, Nigeria [48]. It is surrounded by other high-risk LF towns like Irrua, Uromi and Ubiaja. Ekpoma is estimated to have a growing population of 834,750 with an estimated population growth rate of 3.0 [48]. The town is classified as mainly a semi-urban community with most of the population engaged in small to medium-scale agricultural practices, including hunting and the collection of wild resources [49] (Fig. 1).

4 Sample size determination

The minimum sample size was calculated using the Kish and Leslie formula for estimating a single population proportion [50]: $n = Z^2P\ (1 - P)/d^2$ where $Z = 1.96$ (95% confidence level), $P = 0.78$, based on the reported proportion of households with poor practices associated with Lassa fever in Ondo State, Nigeria [51], and $d = 0.05$. The minimum required sample size was 234 participants. To account for potential non-response, incomplete questionnaires, and ineligible households, 303 participants were recruited for the study.

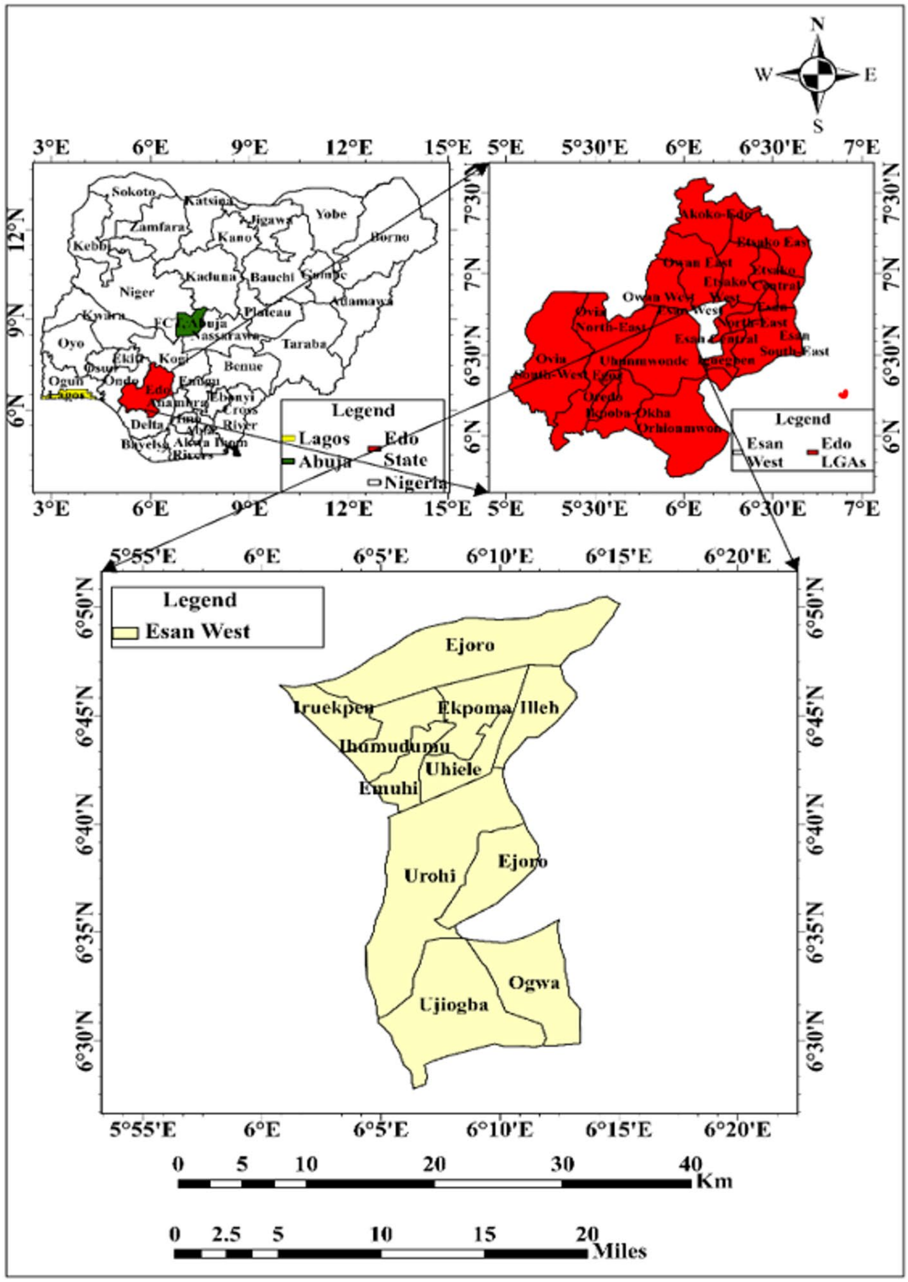

Fig. 1 Map of Nigeria, Edo state and Esan West highlighting the study areas

5 Sampling technique

A multistage sampling technique was employed in the study participant selection. In the first stage, the list of all settlements and villages in and around Ekpoma was obtained from the Esan West LGA authorities. Eighteen of these settlements/villages were randomly selected by balloting. The selected settlements varied in size, with an estimated average of 50–100 households per settlement. In Stage two, households were sampled from each selected settlement using a spatial systematic random sampling with a random start technique. The middle of each settlement was identified, and then a pen was tossed. Sampling started from the house in the direction of the tip of the pen. For settlements with fewer than 20 houses, one house was skipped, while two houses were skipped for settlements with more than 20 houses. Sampling continued until the required sampling size expected for the settlement was attained.

Finally, in Stage three, eligible participants in each selected household who met the inclusion criteria were interviewed. A household was defined as a group of people living together daily in the same house and sharing meals from the same pot [52]. In a scenario where we encountered a compound with multiple independent households sharing a building, each distinct household unit (with its own head and cooking arrangements) was considered a separate sampling unit, and only one household was randomly selected per compound to maintain the independence of observations.

6 Study population

The study participants were heads of households or their representatives from randomly selected settlements in Ekpoma town from June to August 2023.

6.1 Eligibility criteria

6.1.1 Inclusion criteria

A household selected for the study must have resided for at least 2 years in the settlement prior to the study interview. The respondent must be 18 years and above and voluntarily consent to participate in the study.

6.1.2 Exclusion criteria

Selected households were excluded if the head of household or a representative was absent at the time of the household visit to administer the questionnaire, or if they refused to provide informed consent to participate in the study. Additionally, potential respondents with poor cognitive function were also excluded from the study.

6.2 Data collection tools

Data collection was conducted using structured questionnaires for the interviews. The questionnaire was not developed de novo for this study. Rather, it was adapted from previously validated instruments used in similar contexts. The primary sources were: (1) the questionnaire used by Bonwitt et al. [47] for studying rodent hunting and consumption in Sierra Leone; and (2) the instrument developed by Isere et al. [11] for assessing LF preventive practices in Ondo State, Nigeria. Adaptations were made to reflect the specific ecological and cultural context of Ekpoma communities, including the addition of locally relevant food storage items (e.g., garri, fufu, SemoVita flour) and local rodent species names. The adapted questionnaire was then subjected to the validity and reliability assessments described below.

6.3 Pilot testing, reliability, and validity

Prior to the main data collection, a pilot study was conducted in April 2023 among 30 households (approximately 10% of the calculated sample size) in a non-selected community (Ubiaja, Esan South-East LGA) with similar socio-demographic and ecological characteristics to those of Ekpoma. The pilot served to assess questionnaire clarity, comprehension, flow, and average administration time.

Reliability of the questionnaire was assessed using internal consistency measures. The Cronbach's alpha coefficient for the knowledge and practice items was 0.82, indicating good internal consistency. Test-retest reliability was assessed by administering the same questionnaire to 15 pilot respondents after a two-week interval; the intraclass correlation coefficient (ICC) was 0.79 (95% CI: 0.65–0.88), indicating acceptable stability over time.

Content validity was established through review by a panel of five experts: two epidemiologists, one public health physician, one zoonotic disease specialist, and one health promotion expert. The panel assessed each item for relevance, clarity, and comprehensiveness. The Item-Content Validity Index (I-CVI) ranged from 0.80 to 1.00, and the Scale-Content Validity Index (S-CVI) was 0.91, exceeding the acceptable threshold of 0.80. Face validity was confirmed through feedback from pilot respondents, who indicated that the questions were understandable and relevant to their lived experiences.

The final standardised questionnaire was created and deployed in KoboToolbox (http://kobotoolbox.org) and was administered to collect respondents' sociodemographic information, including sex, age, and level of education. Data were also collected on household and behavioural practices that predispose individuals to LF infection, including contact with rodents in homes and farms, contact during hunting activities, and consumption of rodents and food items exposed to rodents. Additional information was obtained on knowledge of LF, as well as bush-burning and food storage practices.

For questions on rodent species identification, study participants were provided with pictorial guides to facilitate accurate identification of the type of rat encountered. These guides display distinguishing morphological features and the local appearance of the multimammate rat alongside other common domestic rodent species. This approach minimised rodent species misclassification.

Operational definitions were provided for key variables: Rodent control measures included trapping, use of rodenticides, keeping cats, or environmental cleaning. Contact with rodents included any physical handling (dead or alive) or direct touching of rodent excreta/urine.

The data was collected using smartphones equipped with KoboCollect software (http://kobotoolbox.org).

6.4 Data analysis

Data collected was analysed using the Statistical Package for the Social Sciences (SPSS) software version 26 (SPSS Inc., Chicago, IL, USA).

6.5 Data management before analysis

Data management procedures included daily backup of raw data to password-protected external storage devices. Data quality was ensured through range and consistency checks to identify out-of-range values and logical inconsistencies. Responses recorded as "Don't know" or left blank were treated as missing data and were excluded from the analysis of the respective variables. Categorical variables were coded numerically to facilitate importation and analysis in IBM SPSS Statistics software. To maintain confidentiality, all datasets were de-identified and securely stored, with access restricted to the principal investigator and the statistician. No data imputation or alteration was performed, and all analyses were conducted using the original verified dataset.

Data virtualisation (electronic processing): In this study, data was collected electronically using KoboCollect software installed on smartphones. The data was then exported directly from the KoboCollect website (http://kobotoolbox.org) as Microsoft Excel files and subsequently imported into SPSS version 26 for cleaning, coding, and statistical analysis.

6.6 Denominator determination for percentages

The denominator used for percentage calculations varied according to the survey structure, including skip patterns, branching logic, and response rates. For variables assessed among all respondents, such as age, sex, and awareness of LF, percentages were calculated using the total sample size ($N=303$). For follow-up questions administered only to a subset of respondents, the denominator comprised only those who met the eligibility criteria for the question. For example, questions on the type of rodent encountered were analysed using only respondents who reported rodent presence in their households ($n=284$). For multiple-response questions, such as knowledge of LF transmission pathways, percentages were calculated using the total sample size as the denominator; consequently, percentages did not sum to 100% because respondents were permitted to select more than one response option.

For the primary outcome variable, reported household LF occurrence, respondents who answered, "Don't know" ($n=41$; 13.5%) were excluded from the logistic regression analysis because outcome status could not be determined. To assess the potential for exclusion bias, the sociodemographic characteristics of excluded and included households were compared using chi-square tests. No statistically significant differences were observed with respect to age ($p=0.342$), sex ($p=0.487$), or educational attainment ($p=0.215$). Sensitivity analyses were subsequently conducted by alternatively classifying "Don't know" responses as "No" and as "Yes." The direction and statistical significance of the principal associations, particularly those relating to food storage on the floor and storage in uncovered containers, remained unchanged, suggesting that exclusion of these responses did not materially influence the study findings. Missing data and "prefer not to answer" responses were excluded from the denominator for the specific variable under analysis.

6.7 Independent and dependent variables

The primary outcome variable was self-reported household LF occurrence within 12 months preceding the survey. This variable was coded as a binary outcome (Yes = 1, No = 0). Respondents who selected "Don't know" were excluded from the regression analysis.

The independent variables were categorised into socio-demographic, knowledge and awareness, and household practice and behavioural factors. Socio-demographic variables included gender, age group, occupation, ethnicity, marital status, religion, highest level of education, number of children, and household size. Knowledge and awareness variables comprise prior awareness of LF, knowledge of LF transmission pathways, and receipt of community sensitisation messages.

Household practices which constituted the primary exposures of interest, included the presence of rodents in or around the household, the type of rodent encountered (multimammate rat or other species), implementation of rodent control measures, physical contact with rodents (dead or alive), physical contact with rodent urine or excreta, hunting rodents for food, food storage practices, consumption of food contaminated by rodent urine or faeces, and engagement in bush-burning practices. Food storage practices were assessed using multiple indicators, including storage in open pots, nylon bags, on shelves, in uncovered containers, in open sacks, and directly on the floor.

The primary outcome was defined as reported prior household LF occurrence by a household member. No independent verification against medical records was performed. In addition, "Don't know" responses ($n = 41$, 13.5%) were excluded from the outcome analysis in the logistic regression model. To assess potential exclusion bias, characteristics of excluded and included households were compared using the chi-square test. No major differences were observed across key sociodemographic variables. Sensitivity analyses were additionally conducted by alternatively classifying "Don't know" responses as "No" and "Yes." Findings remained materially unchanged, suggesting robustness of the primary analysis.

Independent variables for inclusion in the multivariate logistic regression model were selected based on established epidemiological literature, theoretical relevance to LF transmission, and their potential role as confounders, rather than solely on bivariate statistical significance. The following variables were forced into the model regardless of bivariate p-values: age (as a potential confounder), education (as a marker of socioeconomic status and health literacy), household size (as a proxy for crowding and exposure density), and LF exposure of interest (food storage practices: food storage on floor and food storage in uncovered container). Rodent control measures and rodent hunting were also included due to their documented importance in previous studies [11]. This theory-driven approach minimised the risk of omitting important confounders that may not show independent bivariate associations but could modify the exposure-outcome relationship. Results from the final model were reported as adjusted odds ratios (AOR) with corresponding 95% confidence intervals (CIs).

7 Results

The study recruited 303 participants, with more than half being male (58.1%). The mean age of respondents was 44.6 ± 17.8. Over one third of the participants were aged 50 years and above (40.9%), followed by those under 30 (26.4%). Occupations were diverse, with the largest groups being farmers (26.4%) and entrepreneurs (26.1%). The predominant ethnicity was Esan (76.9%), and most participants were married (70.3%) and Christian (95.7%). Regarding education, nearly half had secondary education (47.5%), while 30.4% had completed primary education. Nearly half of the respondents (49.2%) were from households with over five members (49.2%), and more than one third of the respondents were from households with 3–5 children (39.3%), as shown in Table 1.

In Table 2 below, awareness of LF was high, with 92.4% of respondents reporting having heard of the disease. Consumption of food contaminated with rodent urine/faeces was the most frequently identified source of infection (56.6%), followed by contact with LF infected rats (45.4%) and contact with infected people or their bodily fluids (8.6%). Exposure to LF related information through radio broadcasts or interpersonal communication was reported by 76.6% of respondents, while 19.1% reported not having received such information.

Table 3 revealed that most respondents (93.7%) reported having rodents in or around their homes, with 89.1% implementing control measures. Physical contact with rodents was reported by 25.1% of respondents, while 23.8% reported contact with rodent urine

Table 1 Socio-demographic characteristics of the respondents

Variables	Frequency ($N = 303$)	Percentage (%)
Age	80	26.4
< 30	48	15.8
30–39	51	16.8
40–49	124	40.9
50 and above		
Gender	176	58.1
Male	127	41.9
Female		
Occupation	18	5.9
Civil servant	49	16.2
Student	79	26.1
Entrepreneur	80	26.4
Farmer	77	25.4
**Other Occupation		
Ethnicity	233	76.9
Esan	11	3.6
Igbo	59	19.5
*Other tribes		
Marital status	69	22.8
Single	213	70.3
Married	21	6.9
Divorced/single parent/Widowed		
Religion	290	95.7
Christianity	13	4.3
***Other religion		
Highest level of education	20	6.6
No formal education	92	30.4
Primary	144	47.5
Secondary	47	15.5
Tertiary		
How many children do you have	76	25.0
0	41	13.5
1–2	119	39.3
3–5	67	22.1
5+		
Household size	45	14.9
1–2	109	36.0
3–5	149	49.2
5+		

***Other tribes: Etsako, Yoruba, Benin, **Other Occupation: Hunter, Trading and petty shop owners, Artisans, Religious Leaders, ***Other religion: Muslim, Traditional worshippers**

or faeces. Household practices reported by respondents that could facilitate LF exposure were hunting rodents for consumption (37.3%), food storage in uncovered containers (53.8%), and consuming food contaminated with rodent urine or faeces (23.8%). Additionally, 62.4% of respondents reported engaging in bush-burning practices, which may increase contact between humans and rodents.

The association between Sociodemographic characteristics and household practices with household-reported occurrence of LF in the preceding 12 months is presented in Table 4. Households whose respondents had completed only primary education were more likely to report a prior LF occurrence in the preceding 12 months compared with those with post-primary education ($p = 0.019$). Several household practices were also associated with the reported occurrence of Lassa fever in households in the preceding 12 months. Households that implemented rodent control measures reported higher prior LF occurrence compared with those that did not ($p = 0.012$). Similarly, hunting rodents

Table 2 Respondents' knowledge and exposure to Lassa fever

Variable	Frequency	Percentage (%)
Ever heard of Lassa fever		
Yes	280	92.4
No	19	6.3
Don't know	4	1.3
†Knowledge of Lassa fever transmission pathways ("Yes" response reported)		
Consumption of food contaminated with rodent urine/faeces	171	56.4
Contact with Lassa fever–infected rodents	137	45.2
Contact with infected humans	26	8.6
Don't know	95	31.4
‡Other	9	3
Received Lassa fever sensitisation (radio/community messaging)		
Yes	232	76.6
No	58	19.1
Don't know	13	4.3
Household-reported Lassa fever occurrence within 12 months prior to the survey		
Yes	39	12.9
No	223	73.6
Don't know	41	13.5

†Multiple responses were allowed; therefore, percentages do not sum to 100%, ‡Other responses included airborne transmission, mosquito bites, and contact with contaminated hospital equipment.

for food was also associated with reported prior household LF occurrence ($p = 0.028$) and unsafe food storage practices, including food storage in uncovered containers ($p = 0.018$) and food storage on the floor ($p = 0.016$), were associated with reported prior household LF occurrence.

No notable associations were observed for age, household size, presence of rodents around the household, type of rodents encountered, physical contact with rodents or their urine/excreta, consumption of food contaminated by rodents, practised bush burning, or other food storage methods such as food storage in open sacks, nylon bags, pots, or on shelves (Table 4).

Table 5 below presents the multivariate analysis of household practices associated with reported prior household LF occurrence in Ekpoma communities. After adjusting for potential confounders, households where respondents had completed only primary education had higher odds of reported prior household Lassa fever occurrence compared with those who had post-primary education (AOR: 2.63; 95% CI: 1.19–5.79). In addition, households that stored food items on the floor had more than twice the odds of reported prior household Lassa fever occurrence compared with those that did not (AOR: 2.40; 95% CI: 1.11–5.17; $p = 0.026$). Similarly, food storage in uncovered containers was associated with increased odds of reported prior household Lassa fever occurrence (AOR: 5.17; 95% CI: 1.23–21.71; $p = 0.025$).

7.1 Supplementary observation on demographic variation in practices

Unsafe food storage practices varied across demographic groups. Storing food on the floor was more common among farmers (55/80, 68.8%) compared to other occupations, and among respondents with no formal education (9/20, 45.0%) compared to those with

Table 3 Household practices related to rodent exposure among respondents in Ekpoma communities, Edo State, Nigeria ($N=303$)

Variables	Frequency	Percentage (%)
Presence of rodents in or around the household		
Yes	284	93.7
No	19	6.3
***Type of rodent encountered (n = 284)**		
Multimammate rat	27	9.5
******Other rat species	257	90.5
Implementation of rodent control measures within the household and surroundings		
Yes	227	74.9
No	76	25.1
Physical contact with rodents (dead or alive) during the previous 3 months		
Yes	76	25.1
No	227	74.9
Physical contact with rodent urine or excreta		
Yes	72	23.8
No	231	76.2
Hunting rodents for food		
Yes	113	37.3
No	164	54.1
Not respondent, but another household member	26	8.6
Food storage in uncovered pots		
Yes	163	53.8
No	140	46.2
Food storage in open nylon bags		
Yes	3	1
No	300	99
Food storage on open shelves		
Yes	22	7.3
No	281	92.7
Food storage in uncovered containers		
Yes	9	3
No	294	97
Food storage in open sacks		
Yes	13	4.3
No	290	95.7
Food storage on the floor		
Yes	58	19.1
No	245	80.9
Consumed uncooked food contaminated by rodent urine or faeces		
Yes	72	23.8
No	231	76.2
Practice of bush burning		
Yes	189	62.4
No	103	34
Don't know	11	3.6

*For questions with skip patterns (e.g., type of rodent encountered), denominator is indicated as n=284 (respondents who reported rodent presence). For other questions, denominator is N=303

**Other rat species: House rat/Brown rat, Black rat, Bush rat, Giant rat, Grass rat, Food items include Garri, fufu, SemoVita flower, yam powder

Table 4 Sociodemographic and household practices associated with household-reported prior Lassa fever occurrence in Ekpoma communities, Nigeria

Variable	Yes, *n* (%)	No, *n* (%)	Total *N* (%)	χ^2/ Fisher's Exact	*p*-value
Age					
< 40 years	16 (12.5)	112 (87.5)	128 (100.0)	0.027	0.869
≥ 40 years	23 (13.1)	152 (86.9)	175 (100.0)		
Highest education level					
Completed primary education	21 (18.8)	91 (81.2)	112 (100.0)	5.475	0.019*
Post-primary education	18 (9.4)	173 (90.6)	191 (100.0)		
Household size					
< 5 members	9 (11.0)	73 (89.0)	82 (100.0)	0.36	0.548
≥ 5 members	30 (13.6)	191 (86.4)	221 (100.0)		
Presence of rodents in or around the household					
Yes	38 (13.4)	246 (86.6)	284 (100.0)	1.046†	0.485
No	1 (5.3)	18 (94.7)	19 (100.0)		
‡Type of rodent encountered					
Multimammate rat	6 (22.2)	21 (77.8)	27 (100.0)	2.013	0.158
Other rat species	32 (12.5)	225 (87.5)	257 (100.0)		
Rodents control measures implemented					
Yes	39 (14.4)	231 (85.6)	270 (100.0)	5.471†	0.012*
No	0 (0.0)	33 (100.0)	33 (100.0)		
Physical contact with rodents (dead or alive)					
Yes	21 (15.2)	117 (84.8)	138 (100.0)	1.244	0.265
No	18 (10.9)	147 (89.1)	165 (100.0)		
Physical contact with rodent urine or excreta					
Yes	13 (17.8)	60 (82.2)	73 (100.0)	1.622	0.203
No	26 (11.3)	204 (88.7)	230 (100.0)		
Hunting rodents for food					
Yes	24 (17.9)	110 (82.1)	134 (100.0)	4.815	0.028*
No	15 (9.0)	151 (91.0)	166 (100.0)		
Food storage in uncovered pots					
Yes	24 (14.7)	139 (85.3)	163 (100.0)	1.08	0.299
No	15 (10.7)	125 (89.3)	140 (100.0)		
Food storage in open nylon bags					
Yes	1 (33.3)	2 (66.7)	3 (100.0)	1.131†	0.340
No	38 (12.7)	262 (87.3)	300 (100.0)		
Food storage on open shelves					
Yes	2 (9.1)	20 (90.9)	22 (100.0)	0.302†	0.751
No	37 (13.2)	244 (86.8)	281 (100.0)		
Food storage in uncovered containers					
Yes	4 (44.4)	5 (55.6)	9 (100.0)	8.245†	0.018*
No	35 (11.9)	259 (88.1)	294 (100.0)		
Food storage on the floor					
Yes	13 (22.4)	45 (77.6)	58 (100.0)	5.824	0.016*
No	26 (10.6)	219 (89.4)	245 (100.0)		
Consumed food contaminated by rodent urine or faeces					
Yes	11 (15.3)	61 (84.7)	72 (100.0)	0.488	0.485
No	28 (12.1)	203 (87.9)	231 (100.0)		
Food storage in open sacks					
Yes	1 (7.7)	12 (92.3)	13 (100.0)	0.325†	0.483
No	38 (13.1)	252 (86.9)	290 (100.0)		

Table 4 (continued)

Variable	Yes, *n* (%)	No, *n* (%)	Total *N* (%)	χ^2/ Fisher's Exact	*p*-value
‡‡Bush burning practice (n = 292)					
Yes	25 (13.2)	164 (86.8)	189 (100.0)	0.059	0.812
No	14 (13.6)	89 (86.4)	103 (100.0)		

**Level of significance; p < 0.05;*

†Fisher's exact test was reported where one or more expected cell counts were less than 5;

‡Analysis restricted to respondents who reported the presence of rodents in or around the household (n = 284)

‡‡Analysis excluded "Don't know" responses (n = 292)

Table 5 Factors Associated with Reported Household Lassa Fever Occurrence within 12 Months prior to the survey in Ekpoma Communities, Nigeria

Variables	Household reported Lassa fever occurrence within 12 months prior to the survey			
	Unadjusted Odds Ratio (95% CI)	***p*-value**	**Adjusted Odds Ratio (95% CI)**	***p*-value**
Age				
≥ 40 years	1.06 (0.54–2.097	0.869	1.57 (0.69–3.54)	0.280
< 40 years	R			
Highest level of Education				
Completed primary education	2.22 (1.13–4.37)	0.021	2.63 (1.19–5.79)	0. 017+
Post-primary education	R			
Household size				
≥ 5	1.27 (0.58–2.81)	0.549	1.103 (0.53–2.32)	0.796
< 5	R		R	
Rodents control measures in house and surroundings				
Yes	0.74 (0.32–1.69)	0.495	0.722 (0.306–1.700)	0.456
No	R		R	
Hunting rodents for food				
Yes	1.06 (0.53–2.11)	0.872	1.14 (0.56–2.32)	0.716
No	R		R	
Food storage on the floor				
Yes	2.43 (1.16–5.09)	0.018+	2.40 (1.12–5.19)	0.026+
No	R		R	
****Food storage in uncovered containers**				
Yes	5.92 (1.52–23.09)	0.010+	5.16 (1.23–21.71)	0.025+
No	R		R	

R: reference category in the model, +Significant at <0.05

**The association for storing food in uncovered containers (n=9 exposed households; 4 with reported LF occurrence, 44.4%) should be interpreted with caution. The wide confidence interval (95% CI: 1.23–21.71) reflects the instability inherent in estimates derived from small cell sizes. This association should be considered exploratory and hypothesis-generating rather than definitive tertiary education (4/47, 8.5%). These observations suggest that targeted interventions may be warranted for specific subgroups, though formal interaction testing was beyond the scope of this study.

8 Discussion

The socio-demographic profile of the respondents was characterised by a population in which over half were male (58.1%), approximately 41% were aged 50 years and above, and farmers (26.4%) and entrepreneurs (26.1%) constituted the largest occupational groups. These characteristics are epidemiologically relevant, as older adults and individuals engaged in farming may experience sustained exposure to rodent-infested domestic and agricultural environments, thereby potentially increasing opportunities for contact with *Mastomys natalensis*, the primary zoonotic reservoir of LASV [4, 53].

Additionally, a substantial proportion of respondents reported high levels of general LF awareness (92.4%) and community sensitisation (76.6%), with 56.6% correctly identifying contaminated food as a source of infection and 45.4% identifying rats as a potential source of transmission. However, 53.8% reported storing food in open containers, which may represent a knowledge-to-practice gap. Possible explanations could include structural poverty limiting access to rodent-proof storage containers, cultural preferences for traditional storage methods (e.g., open clay pots), perceived low personal risk, and the absence of an enabling environment, such as limited access to affordable rodent-proof storage options. Qualitative studies may help to further explore these barriers [11, 32].

Further exploratory analysis suggests that education level may modify the relationship between knowledge and reported safe storage practices. Among respondents who reported high knowledge (correctly identifying at least two transmission pathways and one prevention method), 85.7% (36/42) of those with tertiary education reported safe food storage (using covered containers or elevated storage), compared with 41.2% (7/17) of those with no formal education. This pattern suggests that knowledge alone may be insufficient to influence behaviour in the absence of enabling resources and infrastructure that often accompany higher educational attainment. Conversely, among respondents with low knowledge, safe storage practices were generally low (12.5% overall), irrespective of educational level. These findings should be interpreted cautiously; however, they may indicate that tailored intervention approaches could be considered for different population subgroups. Formal testing of interaction effects and qualitative exploration of these pathways are recommended for future research.

More importantly, households that stored food items on the floor had more than twice the odds of reported household LF occurrence, while storage of food in uncovered containers was associated with approximately five-fold higher odds of reported household LF occurrence. These findings represent statistical associations and should not be interpreted as causal, given the cross-sectional design. The observed associations are consistent with, but do not establish, the hypothesis that indirect exposure through ingestion of food contaminated with rodent urine or faeces may contribute to LF transmission in endemic settings. Alternative explanations, including reverse causality (where households with prior LF occurrence may have modified their practices following an event) and residual confounding from unmeasured factors such as socioeconomic status and housing conditions, cannot be excluded [54–56].

These findings are consistent with prior community-based studies from Nigeria and Sierra Leone, which reported unsafe food storage practices as being associated with reported household LF occurrence [11, 57, 58]. Storing food on the floor or in uncovered containers plausibly increases rodent access and the likelihood of environmental contamination, even in households reporting rodent control activities. However, this interpretation should be made with caution. The attenuation observed after adjustment may reflect overlap between household food storage and rodent control practices, residual confounding, or other unmeasured household-level factors. Formal assessment of collinearity or interaction was not conducted in this study; therefore, no definitive conclusions can be drawn regarding the interrelationship of these variables. Further studies using longitudinal designs and more robust modelling approaches are needed to better clarify these relationships.

Although hunting rodents for food was associated with reported household LF occurrence in the bivariate analysis, this relationship was attenuated in the multivariable model. Previous studies have identified rodent hunting and consumption as factors associated with LF occurrence in rural West African settings [59, 60]. The absence of an independent association in this study may reflect contextual differences in dominant exposure pathways, with indirect food contamination potentially playing a more prominent role than direct rodent handling. Additionally, the binary classification of hunting behaviour may not have captured variation in frequency, handling, or preparation practices, leading to potential exposure misclassification.

Several behaviours previously implicated in reported household LF occurrence [58, 61–64], including the presence of rodents in or around households, physical contact with rodents or their excreta, reported consumption of contaminated food, and bush burning, were not significantly associated with reported household LF occurrence in this study. This pattern is consistent with observations from hyperendemic settings where rodent exposure is widespread, thereby limiting variability in exposure and reducing statistical power to detect associations [61, 64, 65]. Furthermore, such exposures may be difficult to measure accurately through self-report and may be subject to underreporting, potentially attenuating observed associations [65].

More importantly, these findings identify household food storage practices as important correlates of reported household LF occurrence in this study setting. While unsafe food storage practices were significantly associated with reported household LF occurrence in the adjusted analysis, the cross-sectional design precludes conclusions regarding temporality or causality. The findings should therefore be interpreted as evidence of association rather than causation. Further longitudinal and intervention studies are warranted to better understand these relationships.

The recommendation for widespread adoption of rodent-resistant food storage approaches should be considered due to local feasibility. Potential low-cost and locally appropriate options include improved containers fabricated from recycled materials (e.g., metal drums with tight-fitting lids or modified plastic buckets), raised storage platforms (e.g., elevated wooden shelves or hanging baskets), community-based shared grain storage facilities with rodent-proofing features, and locally fabricated clay pots fitted with secure lids. The feasibility, acceptability, and effectiveness of these options were not assessed in this study and would require further evaluation in partnership with communities.

The implications of these findings for community-led interventions warrant careful consideration. Given the generally high levels of awareness and community sensitisation, future LF prevention efforts may benefit from shifting beyond awareness creation towards supporting the adoption of safer household food storage and environmental management practices. Interventions such as improving access to appropriate food storage containers and promoting coordinated community-wide rodent control strategies may represent potential areas for further investigation. However, the effectiveness of such approaches cannot be inferred from the present study and would require dedicated implementation research.

8.1 Study limitations

The findings of this research should be interpreted with the following limitations in view. First, the cross-sectional nature of the study design implies that the associations identified do not establish a definitive causal relationship. While the reported practices and the reported prior household LF occurrence were both situated within a twelve-month timeframe, the potential for reverse causality remains, as a household that previously experienced LF may have since modified its behaviours due to heightened risk perception.

Second, the reliance on self-reported data introduces the possibility of recall bias, particularly concerning events that occurred within a year prior the interview. There is also a risk of social desirability bias, where participants might under-report practices known to predispose to LF disease (such as rodent hunting) or over-report the effectiveness of their rodent control measures to align with public health messaging.

Third, outcome misclassification is a significant concern. The primary outcome was self-reported household LF occurrence based on respondent recall. No independent verification against medical records or surveillance data was performed. Respondents may have confused LF with other febrile illnesses (e.g., malaria, typhoid fever, other viral haemorrhagic fevers), leading to potential over-reporting. Conversely, mild or asymptomatic cases may have been missed, leading to under-reporting. If misclassification were non-differential with respect to exposure, associations would be biased towards the null. However, differential misclassification (e.g., households with known LF occurrence more accurately recalling exposures) cannot be ruled out.

Fourth, the p-value-driven variable selection approach (including only variables significant at the bivariate level in the multivariate model) risks omitting important confounders that should be included a priori based on epidemiological reasoning. Important confounders such as socioeconomic status (beyond education as a proxy), precise household construction quality (e.g., presence of ceiling, wall type, floor type), and detailed wealth index were not forced into the model. Residual confounding from these unmeasured factors remains possible.

A further limitation concerns the small sample sizes for certain exposure categories. Only nine households reported storing food in containers without covers, of which four reported prior LF occurrence. While the association was statistically significant ($p = 0.025$) and the odds ratio was large (AOR = 5.17), the wide confidence interval (1.23–21.71) indicates substantial uncertainty. Such sparse data estimates can be unstable and may provide a different outcome in larger samples. These findings should therefore be regarded as suggestive and in need of confirmation through larger, adequately powered studies.

Finally, while Ekpoma communities in Edo State represent a critical hyperendemic hotspot for LF in Nigeria, the socio-cultural and ecological factors specific to this region may not be perfectly generalisable to other endemic areas in Nigeria or the wider West African sub-region that possess different environmental conditions or behavioural norms. Future research utilising a longitudinal design or multi-site comparative approach would be necessary to establish a more broadly applicable and causal risk profile.

9 Conclusion

The findings highlight associative patterns between socio-demographic factors, knowledge, and household practices with reported prior household LF occurrence in endemic communities in Nigeria. While community awareness of LF was high, household practices such as storing food on the floor and using uncovered containers were associated with reported prior household LF occurrence. Although rodent control measures were not independently associated with reported household LF occurrence after adjustment, this should not be interpreted as evidence of ineffectiveness, as the observed attenuation may reflect overlap with other household practices or residual confounding. The findings nevertheless suggest that household environmental conditions and food storage practices warrant further investigation as potential targets for LF prevention in hyperendemic settings. Consequently, interventions may benefit from complementing health education with community-based environmental and structural approaches, including improved food storage practices and coordinated rodent control activities. Future research employing longitudinal designs, formal assessment of interaction between household practices, qualitative approaches to explore barriers to behavioural change, and multi-site comparative studies would help clarify these relationships, establish temporality, and assess generalisability.

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Author contributions
JOI conceived the study; JOI and EEI designed the study protocol, JOI and AA carried out the data collection, EEI and OA carried out analysis and interpretation of data , JOI and EEI drafted the manuscript; JOI, EEI, DA and AO critically revised the manuscript for intellectual content. All authors read and approved the final manuscript. JOI and EEI are guarantors of the paper.

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Data availability
The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. The corresponding author could be contacted through [joigbokwe@oauife.edu.ng](mailto: joigbokwe@oauife.edu.ng) .

Declarations

Ethics approval and consent to participate
The study protocol was reviewed and approved by the Irrua Specialist Hospital Ethics Review Committee, Irrua, Edo State, Nigeria, with protocol approval number ISTH/HREC/20230202/444 in accordance with the Declaration of Helsinki and ICH-GCP guidelines. Informed consent was obtained from all respondents. Before data collection, respondents were approached individually by trained research assistants and given a detailed explanation of the study. The research

assistants introduced themselves, explained the study's objectives, and ensured that each respondent understood the purpose of their participation. The explanation was provided in a language (English, Pidgin) that respondents were comfortable with to enhance comprehension. Respondents were informed that participation was entirely voluntary, meaning they had the right to refuse or withdraw at any time without facing any consequences. They were reassured that their decision would not affect their access to healthcare services or other benefits of the study. To formalise consent, respondents were provided with an Informed Consent Form outlining the study's purpose, procedures, potential risks, benefits, and confidentiality measures. If literate, they were asked to read and sign the form. If illiterate, a witness who could read and write was asked to interpret the content of the form and explain it to the understanding of the respondent, after which the respondent provided consent through a thumbprint, witnessed by the interpreter. Signed or thumbprinted forms were securely stored as proof of consent before proceeding with data collection. Data collected from respondents was kept confidential using a password-protected Kobocollect database with access provided only to the principal investigator. No monetary compensation was provided to participants.

Consent for publication
Not Applicable.

Competing interests
The authors declare no competing interests.

Dual publication
The authors confirm that this work has not been published previously and is not under consideration for publication elsewhere.

Authorship
All authors meet the ICMJE criteria for authorship. JOI conceived the study; JOI and EEI designed the study protocol; JOI and AA carried out the data collection; EEI and OA carried out the analysis and interpretation of data; JOI and EEI drafted the manuscript; JOI, EEI, DA and AO critically revised the manuscript for intellectual content. All authors read and approved the final manuscript. JOI and EEI are guarantors of the paper.

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